Jon Roberts

Minneapolis / St. Paul, MN
jon@precursive.dev
(512) 828-7363
github.com/emptyflask
linkedin.com/in/emptyflask

summary

Lead software engineer with 15+ years building Ruby on Rails and, more recently, TypeScript systems. I specialize in search, authentication, and high-traffic event platforms, with Nix at work and at home.

skills

Languages	Ruby, TypeScript, JavaScript, Haskell, SQL, PHP
Frameworks	Ruby on Rails, Hanami, React, SolidJS
Search	OpenSearch / Elasticsearch
Auth	SAML, OAuth2 / PKCE, JWT, SSO, Okta, Doorkeeper
Data	PostgreSQL, pgvector, Redis, MySQL, CouchDB
Infra	Nix, Docker, Ansible, AWS, Github Actions
OS	Linux (NixOS, Debian / Ubuntu, Gentoo), macOS

experience

Mar 2022 – Present — **Lead Software Engineer** — Remote

South by Southwest

- Technical lead for a team of 6–10 developers. Build reference architecture the team reuses across projects, review pull requests, and mentor less experienced engineers.
- Designed a search architecture for the SXSW schedule, using OpenSearch (Elasticsearch) as a read-optimized store for the frontend and Postgres as the normalized source of truth.
- Built custom index mappings and denormalized documents across several models, with a presentation layer that shapes results, highlights, and aggregations for faceted search in a React frontend.
- Led a phased migration to unified SSO across SXSW's Rails and WordPress properties: Clearance sessions with Redis broadcast, a bridge between the .sxsw.com and .sxswedu.com domains, and OAuth2 / PKCE with JWT for the mobile app.
- Rebuilt the attendee recommendations system on pgvector.
- Run development workflows in Jira and GitHub, maintain CI pipelines, Docker and Ansible configuration, and Nix dev environments, and handle AWS infrastructure changes.
- Building MCP tools that let attendees query the schedule through LLM agents.

Mar 2021 – Mar 2022 — **Senior Software Engineer** — Remote

Proof (formerly Notarize)

- Built SAML single sign-on through Okta for internal staff, mapping Okta groups to role-based access control in the app.
- Integrated with eOriginal's eVault to securely store the authoritative copies of digitally signed documents.
- Built a CLI tool around a Java PDF library for document processing.
- Refactored flaky and slow rspec+factorybot tests to improve CI reliability.
- Converted / refactored legacy JavaScript to TypeScript.
- Migrated application configuration to AWS Parameter Store as part of the move from Heroku to AWS.

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Oct 2012 – Mar 2021 — **Senior Software Engineer** — Austin, TX
South by Southwest

- Owned SXSW's core attendee systems: the schedule platform (schedule.sxsw.com), the e-commerce cart and checkout, and the central OAuth / SSO provider used across SXSW's apps.
- Built cart checkout, coupon redemption, guest-to-user cart merging, and payment processing with ActiveMerchant, plus a coupon management service and a centralized transaction-logging API for payment gateways.
- Built PanelPicker's public voting and scoring system for session proposals, and messaging, badges, and Facebook login for the SXSW social app.
- Built on-site registration tools: wristband and badge distribution with registration import and deduplication, FileMaker-integrated artist registration, and a volunteer badge printing app using the Shiftboard API.
- Built an image hosting and manipulation service on Thumbor, with client gems that integrated it into SXSW's Rails apps.

Sep 2009 – Oct 2012 — **Independent Contractor** — Austin, TX
Various

- Built Rails, Drupal, and WordPress projects for clients including SXSW, Dell, UT Austin, Austin Humane Society, and Scott & White.

Sep 2008 – Sep 2009 — **Lead Developer** — Austin, TX
Pixeldust Interactive

- Built a 12-language, PDF-generating Drupal site for Johnson Controls, plus CakePHP and WordPress sites for other clients.

1998 – 2008 — **Earlier roles**
Web / UX designer, and CG artist roles in Austin and Indianapolis.

education

1999 – 2003 — **BFA in Computer Art** — Savannah, GA
Savannah College of Art & Design
Studied VFX, compositing, procedural effects, shader programming, design, and a ton of art history.

projects

nix-config — Multi-host NixOS flake for my desktops and homelab.
rails-nix — Example Rails app using Nix for all dependencies.
other open-source — Contributed to nixpkgs, dry-rb, elasticsearch-rails, xmonad-contrib, fmrest-ruby, etc.

interests

Functional programming (Haskell, Elixir, Gleam, Elm, Rust*), cooking, gardening, mushroom foraging, guitar (badly), bicycles, photography.